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### SAP S/4 HANA: Simplifications in Sales & Distribution Data Models

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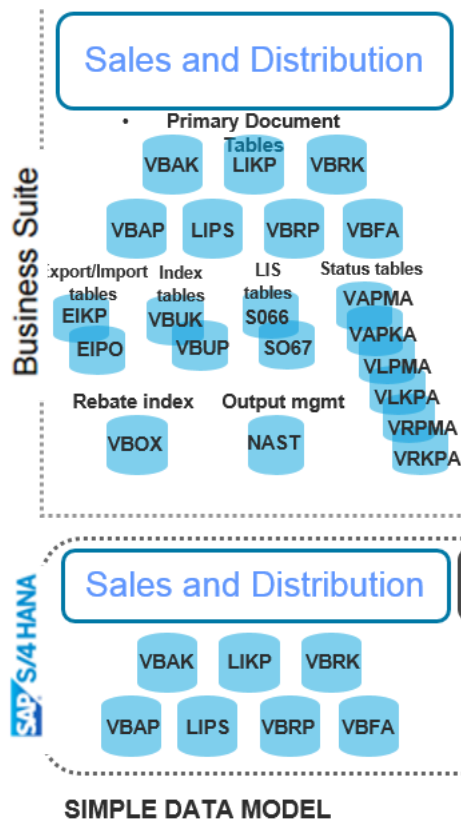
SAP S/4 HANA is the new offering from SAP built on the high performance In-memory platform HANA with an enriching user experience using FIORI apps. This new system includes major changes and massive simplifications ranging from changes in underlying data models to new user interface through FIORI apps.

#### Objective:

The objective of this blog is to understand the data model simplifications in SD area through a comparative study with a non S/4 system.

Below are the major simplification points:

- Status tables VBUK and VBUP have been eliminated and the new status fields have been added to:
  - VBAK and VBAP for sales order header and item
  - LIKP and LIPS for delivery document header and item
  - VBRK for billing document header
  - No status fields are present for billing items.
- Simplification of document flow table VBFA
- Document index tables like VAKPA, VAPMA etc. have been done away with.



## Comparison:

Let us understand the differences by looking at the table structures:

- VBUK and VBUP are still present in S/4 HANA but they are not filled when an order is created.
- The status fields have been added through append structure.

| Table Name | Description             | Append Name |
|------------|-------------------------|-------------|
| VBAK       | Sales Order Header      | VBAK_STATUS |
| VBAP       | Sales Order Item        | VBAP_STATUS |
| LIKP       | Delivery Header         | LIKP_STATUS |
| LIPS       | Delivery Item           | LIPS_STATUS |
| VBRK       | Billing Document Header | VBRK_STATUS |

- If any custom fields have been added to VBUP or VBUP tables in source system, they will have to be added to respective document tables in S/4 HANA system.
- An append field to the document header status table VBUK must be added to one or several of the document header tables VBAK, LIKP, or VBRK. This decision depends on which of the document types the respective field is relevant for.
- An append field to document item status table VBUP must be added to one or more of the document item tables VBAP or LIPS.

## VBAK:

**Dictionary: Display Table**

Transparent Table: VBAK, Active

Short Description: Sales Document: Header Data

Attributes: Delivery and Maintenance, Fields, Input Help/Check, Currency/Quantity Fields

| Field   | Key                      | Table | Data element | Data Type | Length | Decim. | Short Description                                         | Group       |
|---------|--------------------------|-------|--------------|-----------|--------|--------|-----------------------------------------------------------|-------------|
| INCLUDE | <input type="checkbox"/> | VBAK  | STATUS       | CHAR      | 0      |        | Status include for sales order document header (VBAK)     | VBAK_STATUS |
| ABSTX   | <input type="checkbox"/> | VBAK  | STATUS       | CHAR      | 1      |        | Overall rejection status of all document items            |             |
| BESTX   | <input type="checkbox"/> | VBAK  | STATUS       | CHAR      | 1      |        | Confirmation status                                       |             |
| CMPSD   | <input type="checkbox"/> | CMPSD | CHAR         | 1         |        |        | Status of credit check against maximum document value     |             |
| CMPSD   | <input type="checkbox"/> | CMPSD | CHAR         | 1         |        |        | Status of credit check against terms of payment           |             |
| CMPSI   | <input type="checkbox"/> | CMPSI | CHAR         | 1         |        |        | Status of credit check against financial document         |             |
| CMPSI   | <input type="checkbox"/> | CMPSI | CHAR         | 1         |        |        | Status of credit check against export credit insurance    |             |
| CMPSI   | <input type="checkbox"/> | CMPSI | CHAR         | 1         |        |        | Status of credit check against payment card authorization |             |
| CMPSI   | <input type="checkbox"/> | CMPSI | CHAR         | 1         |        |        | Status of Credit Check SAP Credit Management              |             |
| CMPSI   | <input type="checkbox"/> | CMPSI | CHAR         | 1         |        |        | Status of Technical Error SAP Credit Management           |             |
| CMPSI   | <input type="checkbox"/> | CMPSI | CHAR         | 1         |        |        | Overall status of credit checks                           |             |

## VBAP:

**Dictionary: Display Table**

Transparent Table VBAP Sales Document: Item Data Active

Short Description:

Attributes:  Delivery and Maintenance  Fields  Input Help/Check  Currency/Quantity Fields

Field  Key  Data element  Data Type  Length  Decim...  Short Description  236 / 377  Group

| Field   | Key                      | Data element | Data Type | Length | Decim... | Short Description                                      | Group       |
|---------|--------------------------|--------------|-----------|--------|----------|--------------------------------------------------------|-------------|
| INCLUDE | <input type="checkbox"/> | VBAP_STATUS  | STRU      | 0      |          | Status include for sales order document item (VBAP)    | VBAP_STATUS |
| ABSTA   | <input type="checkbox"/> | ABSTA_VB     | CHAR      | 1      |          | Rejection status for SD item                           |             |
| BESTA   | <input type="checkbox"/> | BESTA        | CHAR      | 1      |          | Confirmation status of Document Item                   |             |
| CMPE1   | <input type="checkbox"/> | CMPE1        | CHAR      | 1      |          | Status of credit check against financial document      |             |
| CMPE2   | <input type="checkbox"/> | CMPE2        | CHAR      | 1      |          | Status of credit check against export credit insurance |             |
| COSTA   | <input type="checkbox"/> | COSTA_D      | CHAR      | 1      |          | Confirmation status for ALE                            |             |
| DCSTA   | <input type="checkbox"/> | DCSTA        | CHAR      | 1      |          | Delay status                                           |             |
| FSABA   | <input type="checkbox"/> | FSABA        | CHAR      | 1      |          | Billing Status for Order-Related Billing Documents     |             |
| FSSTA   | <input type="checkbox"/> | FSSTA        | CHAR      | 1      |          | Billing block status for items                         |             |

When a sales order is created in a non S/4 HANA system:

Sales Order: 7137:

**Display Std Order-w shipment 7137: Overview**

Std Order-w shipment: 7137 Net value: 200,00 USD

Sold-To Party: 100000

Ship-To Party: 100000

PO Number: PO date:

Item overview Item detail Ordering party Procurement Shipping Reason for rejection Fast Entry Screen

General header data

Description:

Req. deliv. date: 01.06.2016 Deliver. Plant:

Complete div.: ☐ Total Weight: 2 LB

Delivery block:  Volume: 0,000

Billing block:  Pricing date: 01.06.2016

Total amount: 200,00 Doc. Currency: USD / 1,00000

Payment card:  Exp. date:

All items

| Item | Material | Order Quantity | Un | S | Description    | Customer Material Numb | ItCa | DGIP | HL Itm | D First date     | Print | Batch |
|------|----------|----------------|----|---|----------------|------------------------|------|------|--------|------------------|-------|-------|
| 10   | 111A     | 2 EA           |    |   | Checked Shirts |                        | TAZ  |      |        | 01.06.2016 1000  |       |       |
| 20   | 111A     | 2 EA           |    |   | Checked Shirts |                        | TAZS |      |        | 101.06.2016 1000 |       |       |

VBAP table holds the order header details

**Data Browser: Table VBAP Select Entries 1**

Check Table...

| MANDT | VBELN      | ERDAT      | ERZET    | ANGDT | BNDDT | AUDAT      | VBTYP | TRVOC | ALJAR | AUGUR | GWLDT | SUBME | LFSK | FAKSK | NETWR  | WAERK | VKORG | VTVWG | SPART | VKGRP | VKBUR | GSBER | GSKST | GLUEBG | GREEN |
|-------|------------|------------|----------|-------|-------|------------|-------|-------|-------|-------|-------|-------|------|-------|--------|-------|-------|-------|-------|-------|-------|-------|-------|--------|-------|
| 100   | 0000007137 | 01.06.2016 | 04:18:21 |       |       | 01.06.2016 | C     | D     | ZOR   |       |       |       |      |       | 200,00 | USD   | 1000  | 10    | 00    |       |       |       |       |        |       |

Order status information is present in VBUK table

<

Sales Order in S/4 HANA system:

**Display Standard Order 1: Overview**

Standard Order: 1 Net value: 1.000,00 USD

Sold-To Party: 2

Ship-To Party: 2

PO Number: 1234 PO date:

Item overview Item detail Ordering party Procurement Shipping Fast data entry Reason for rejection

Req. deliv. date: 22.06.2016 Deliver. Plant:

Complete div.: ☐ Total Weight: 5 KG

Delivery block:  Volume: 0,000

Billing block:  Pricing date: 22.06.2016

Payment card:  Exp. date:

Card Verif. Code:

Payment terms: 0001 Pay immediately w/o deduction

Inco. Version:

All items

| Item | Material | Order Quantity | Un | S | Description         | Customer Material Numb | ItCa | DGIP | HL Itm | D  |
|------|----------|----------------|----|---|---------------------|------------------------|------|------|--------|----|
| 10   | 2009     | 1 KG           |    |   | RAW09,PO,Batch-Fifo |                        | TAZ  |      |        | 01 |

VBAP table contains both the header details as well as the header status

**Data Browser: Table VBAP Select Entries 1**

Check Table...

| MANDT | VBELN      | ERDAT      | ERZET    | ANGDT | BNDDT | AUDAT      | VBTYP | TRVOC | ALJAR | AUGUR | GWLDT | SUBME | LIF | FAKSK | NETWR    | WAERK | VKORG | VTVWG | SPART | VKGRP | VKBUR | GSBE | GLUEB | GLUEE |
|-------|------------|------------|----------|-------|-------|------------|-------|-------|-------|-------|-------|-------|-----|-------|----------|-------|-------|-------|-------|-------|-------|------|-------|-------|
| 200   | 0000000001 | 22.06.2016 | 02:05:31 |       |       | 22.06.2016 | C     | 0     | TA    |       |       |       |     |       | 1.000,00 | USD   | 1710  | 10    | 00    | 170   |       |      |       |       |

Header status fields are now added to VBAP table

**Data Browser: Table VBAP Select Entries 1**

Check Table...

| MANDT | VBELN      | FKSAK | FMST | FSSTK | GBST | LFSGK | LFSTK | LSSTK | MAN | RFSGK | RFSTK | SPSTG | TRST | UVALL | UVALS | UVFA | UVFAS | UVPR | UVVL | UVVLS |
|-------|------------|-------|------|-------|------|-------|-------|-------|-----|-------|-------|-------|------|-------|-------|------|-------|------|------|-------|
| 200   | 0000000001 |       |      |       |      |       |       |       |     |       |       |       |      |       |       |      |       |      |      |       |

Table VBUK is present but is not filled.

**Data Browser: Table VBUK: Selection Screen**

Number of Entries

VBELN  to

Width of Output List

Maximum No. of Hits

☒ No table entries found for specified key

## Reason VBUK / VBUP are still present in S/4 HANA:

Since the status tables are not being filled it can be questioned why the tables are still present and not removed from system altogether. The reason for this is to enable smooth transition for those opting for migration to S/4 HANA system.

New function modules have been written which read the document status fields from VBAK , LIKP aur VBRK tables and populate in output whose structure is similar to that of VBUK.

For example FM SD\_VBUK\_READ\_FROM\_DOC is a new FM that fills VBUK structure for one SD document out of document header tables. The data is fetched depending upon the document type.

**Function Builder: Display SD\_VBUK\_READ\_FROM\_DOC**

Function module: SD\_VBUK\_READ\_FROM\_DOC Active

```

50 lt_field = VALUE #( ( 'MANDT' ) ( 'VBELN' ) ( 'VBTYPE' ) ).
51
52 IF i_vbtyp IS INITIAL AND i_vbobj IS INITIAL.
53 * LOGIC IF TABLE IS NOT KNOWN
54
55 "select with field list is faster than select * only for successful reads, so do select * and save the descriptor access for this read try
56 SELECT SINGLE * FROM vbak INTO CORRESPONDING FIELDS OF es_vbuk WHERE vbeln = i_vbeln.
57 IF sy-subrc EQ 0.
58   es_vbuk-vbobj = 'A'.
59   es_vbukvb = es_vbuk.
60   RETURN.
61   ENDIF.
62
63 "select with field list is faster than select * only for successful reads, so do select * and save the descriptor access for this read try
64 SELECT SINGLE * FROM likp INTO CORRESPONDING FIELDS OF es_vbuk WHERE vbeln = i_vbeln.
65 IF sy-subrc EQ 0.
66   es_vbuk-vbobj = 'L'.
67   es_vbukvb = es_vbuk.
68   RETURN.
69   ENDIF.
70
71 lt_status_field = cl_ad_ddic_util=>get_ddic_field_list_struct_mmp( lc_vbrk_status_incl ).
72 ASSIGN <des of lt_status_field> TO lt_field.
73 SELECT SINGLE (lt_field) FROM vbrk INTO CORRESPONDING FIELDS OF es_vbuk WHERE vbeln = i_vbeln.
74 IF sy-subrc EQ 0.
75   es_vbuk-vbobj = 'F'.
76   es_vbukvb = es_vbuk.
77   RETURN.
78 ELSE.
79   RAISE vbeln_not_found.
80   ENDIF.
81

```

- The output structure ES\_VBUK still refers to VBUK table.
- Any custom fields added to appends like VBAK\_STATUS will be read by this FM.

## Advantages:

- Reduced memory footprint : Since fewer tables are involved this means reduction of redundant data and simplified document flow.
- Now if we need to query the order table on the basis of document status we can simply do so by 1 single query on VBAK table instead of join on VBAK and VBUK.

## NOTES:

- OSS Note 2198647 can be referred for further information.

## Acknowledgment:

- Reference taken from Simplification List for S/4 HANA 1511 0302.

Suggestions and Comments Welcomed !!

~Tanmay

## 22 Comments



Former Member



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Nice. Thanks for sharing :smile: